Overview of Receiver operating characteristic

Subject: Receiver operating characteristic

1.

Probabilistic interpretation The area under the curve (often referred to as simply the AUC) is equal to the probability that a classifier will rank a randomly chosen positive instance higher than a randomly chosen negative one (assuming 'positive' ranks higher than 'negative'). In other words, when given one randomly selected positive instance and one randomly selected negative instance, AUC is the probability that the classifier will be able to tell which one is which. This can be seen as follows: the area under the curve is given by (the integral boundaries are reversed as large threshold T {\displaystyle T} has a lower value on the x-axis)

2.

History The ROC curve was first used during World War II for the analysis of radar signals before it was employed in signal detection theory. Following the attack on Pearl Harbor in 1941, the United States military began new research to increase the prediction of correctly detected Japanese aircraft from their radar signals. For these purposes they measured the ability of a radar receiver operator to make these important distinctions, which was called the Receiver Operating Characteristic. In the 1950s, ROC curves were employed in psychophysics to assess human (and occasionally non-human animal) detection of weak signals. In medicine, ROC analysis has been extensively used in the evaluation of diagnostic tests. ROC curves are also used extensively in epidemiology and medical research and are frequently mentioned in conjunction with evidence-based medicine. In radiology, ROC analysis is a common technique to evaluate new radiology techniques. In the social sciences, ROC analysis is often called the ROC Accuracy Ratio, a common technique for judging the accuracy of default probability models. ROC curves are widely used in laboratory medicine to assess the diagnostic accuracy of a test, to choose the optimal cut-off of a test and to compare diagnostic accuracy of several tests. ROC curves also proved useful for the evaluation of machine learning techniques. The first application of ROC in machine learning was by Spackman who demonstrated the value of ROC curves in comparing and evaluating different classification algorithms. ROC curves are also used in verification of forecasts in meteorology.

3.

where X 1 {\displaystyle X_{1}} is the score for a positive instance and X 0 {\displaystyle X_{0}} is the score for a negative instance, and f 0 {\displaystyle f_{0}} and f 1 {\displaystyle f_{1}} are probability densities as defined in previous section. If X 0 {\displaystyle X_{0}} and X 1 {\displaystyle X_{1}} follows two Gaussian distributions, then A = Φ ((μ 1 − μ 0) / σ 1 2 + σ 0 2) {\displaystyle A=\Phi \left((\mu _{1}-\mu _{0})/{\sqrt {\sigma _{1}^{2}+\sigma _{0}^{2}}}\right)} .

4.

A = ∫ x = 0 1 TPR (FPR − 1 (x)) d x = ∫ ∞ − ∞ TPR (T) FPR ′ (T) d T = ∫ − ∞ ∞ ∫ − ∞ ∞ I (T ′ ≥ T) f 1 (T ′) f 0 (T) d T ′ d T = P (X 1 ≥ X 0) {\displaystyle {\begin{aligned}A&=\int _{x=0}^{1}{\mbox{TPR}}({\mbox{FPR}}^{-1}(x))\,dx\\[5pt]&=\int _{\infty }^{-\infty }{\mbox{TPR}}(T){\mbox{FPR}}'(T)\,dT\\[5pt]&=\int _{-\infty }^{\infty }\int _{-\infty }^{\infty

}I(T'\geq T)f_{1}(T')f_{0}(T)\,dT'\,dT=P(X_{1}\geq X_{0})\end{aligned}}}

5.

Radar in detail As mentioned ROC curves are critical to radar operation and theory. The signals received at a receiver station, as reflected by a target, are often of very low energy, in comparison to the noise floor. The ratio of signal to noise is an important metric when determining if a target will be detected. This signal to noise ratio is directly correlated to the receiver operating characteristics of the whole radar system, which is used to quantify the ability of a radar system. Consider the development of a radar system. A specification for the abilities of the system may be provided in terms of probability of detect, P D {\displaystyle P_{D}} , with a certain tolerance for false alarms, P F A {\displaystyle P_{FA}} . A simplified approximation of the required signal to noise ratio at the receiver station can be calculated by solving

6.

An alternative to the ROC curve is the detection error tradeoff (DET) graph, which plots the false negative rate (missed detections) vs. the false positive rate (false alarms) on non-linearly transformed x- and y-axes. The transformation function is the quantile function of the normal distribution, i.e., the inverse of the cumulative normal distribution. It is, in fact, the same transformation as zROC, below, except that the complement of the hit rate, the miss rate or false negative rate, is used. This alternative spends more graph area on the region of interest. Most of the ROC area is of little interest; one primarily cares about the region tight against the y-axis and the top left corner – which, because of using miss rate instead of its complement, the hit rate, is the lower left corner in a DET plot. Furthermore, DET graphs have the useful property of linearity and a linear threshold behavior for normal distributions. The DET plot is used extensively in the automatic speaker recognition community, where the name DET was first used. The analysis of the ROC performance in graphs with this warping of the axes was used by psychologists in perception studies halfway through the 20th century, where this was dubbed "double probability paper".

7.

Area under the curve It can be shown that the AUC is closely related to the Mann–Whitney U, which tests whether positives are ranked higher than negatives. For a predictor f {\textstyle f} , an unbiased estimator of its AUC can be expressed by the following Wilcoxon-Mann-Whitney statistic:

8.

where 1 [f (t 0) < f (t 1)] {\textstyle {\textbf {1}}[f(t_{0})<f(t_{1})]} denotes an indicator function which returns 1 if f (t 0) < f (t 1) {\displaystyle f(t_{0})<f(t_{1})} otherwise return 0; D 0 {\displaystyle {\mathcal {D}}^{0}} is the set of negative examples, and D 1 {\displaystyle {\mathcal {D}}^{1}} is the set of positive examples. In the context of credit scoring, a rescaled version of AUC is often used: